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Sistem Information Retrieval Berbasis BM25 pada Platform OpenMRS

LAPORAN AKHIR PROYEK STBI

Ade Dwi Prayitno	25/563076/PPA/07097
Aldi Indrawan	25/557923/PPA/07038
Ari Rudiantama	25/569437/PPA/07170
Dimas Ihdam Maulana	25/562999/PPA/07090
Wiladahtul Awaliah	25/569610/PPA/07174
Magister Ilmu Komputer - Universitas Gadjah Mada	

Ringkasan Proyek

Dari proposal sampai sistem berjalan, termasuk ablation study lima varian.



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Masalah & tujuan

- ▶ Pencarian RME *exact-match* gagal menangani sinonim, singkatan, dan variasi terminologi medis.
- ▶ Tujuan: modul IR berbasis BM25 di OpenMRS yang menangkap relevansi semantik teks klinis.
- ▶ Dataset: MTSamples (~5.000 transkripsi, 40 spesialisasi).

Kontribusi

- ▶ Bukan sekadar “BM25 di dataset medis”: studi **ablasi** lapisan normalisasi + ekspansi klinis.
- ▶ 5 varian retrieval (V0-V4) dibandingkan berdampingan.
- ▶ Modul OpenMRS + endpoint JSON + UI dropdown, teruji berjalan.
- ▶ Evaluasi TREC-style: P@K, Recall@K, nDCG@K.



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Progress: Data & Korpus



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Dataset Diverifikasi, Korpus Final Dibangun



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Data mentah diproses menjadi korpus siap indeks.

4.999

baris mentah

2.609

baris duplikat

2.316

dokumen final

91,2%

multi-spesialisasi

Apa yang dikerjakan

- ▶ Integritas CSV diverifikasi byte-for-byte terhadap statistik resmi Kaggle.
- ▶ Pipeline `tools/build_corpus.py`: bersihkan, gabungkan duplikat, saring <50 token.
- ▶ Duplikat **digabung** (bukan dibuang) agar metadata multi-label dipertahankan.

Temuan penting

medical_specialty multi-label (91,2%), **tidak** valid sebagai *ground-truth* relevansi $\Rightarrow$ qrels disusun terpisah (semi-otomatis, sadar-sinonim).

Sumber: `dataset/README.md`, `corpus_stats.txt` [1].



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Kajian Pustaka & Gap



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BM25 pada MTSamples Sudah Diuji, tapi Sebagai Baseline Naif



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Gap: lapisan normalisasi & ekspansi klinis untuk BM25 belum disasar.

Studi	Fokus	Perlakuan BM25
Multi-Corpus Benchmark Study (Mikkelsen, 2026), termasuk MTSamples ($n=500$)	Membandingkan model <i>embedding</i> vs. BM25 untuk retrieval klinis	BM25 hanya <i>baseline</i> : tokenisasi <i>lowercase</i> + hapus tanda baca ($k_1=1,5$, $b=0,75$)
Improved BM25 untuk <i>Precision Medicine</i> (Zhang, 2021)	Optimasi bobot skor BM25 (<i>co-word</i> + Cuckoo Search)	Perbaikan fungsi skor, bukan tokenisasi/normalisasi
Gayen dkk. (2023): Essie $\rightarrow$ Solr	Tokenisasi klinis di Solr	Bukan untuk BM25 di RME
Griffon dkk. (2012): UMLS synonyms	Ekspansi sinonim UMLS	Untuk PubMed, bukan RME

Gap yang diisi

Belum ada studi yang mengevaluasi **lapisan normalisasi + ekspansi sinonim/singkatan klinis** untuk **BM25** pada **MTSamples**, apalagi yang diintegrasikan langsung ke **OpenMRS**.

Sumber: Mikkelsen (2026) [2]; Zhang (2021) [3]; Gayen dkk. (2023) [4]; Griffon dkk. (2012) [5].

Urgensi Diukur Langsung pada Korpus (Bukan Asumsi)



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Lexicon klinis yang dibangun + cakupannya pada 2.316 dokumen.

Negasi klinis

88,0% dokumen (13 term).

Stopword list umum membuang "no/denies/without", padahal maknanya berlawanan.

Istilah majemuk

60,9% dokumen (28 frasa).

Tokenizer kata-tunggal memecah "shortness of breath" jadi token lepas.

Singkatan medis

28,8% dokumen (28 singkatan).

"c/o", "DM", "HTN", "CKD" tak bermakna bila dipecah generik.

Implikasi

Baseline sebelumnya (lowercase + hapus tanda baca) membuang sinyal yang muncul di sebagian besar dokumen. Solusi: normalisasi klinis (negasi dipertahankan, majemuk dikolaps, singkatan diekspansi) + ekspansi sinonim, dibandingkan ablation terhadap baseline naif.

Pengukuran pada `corpus.jsonl` (2.316 dokumen) via `tools/build_lexicon.py`.



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Implementasi



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Lima Varian Retrieval (Ablation Study)



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Parameter BM25 sama ($k_1=1,2$, $b=0,75$); bedanya hanya lapisan tokenisasi.

ID	Label	Strategi
V0	Naive BM25	lowercase + hapus tanda baca (baseline)
V1	Normalisasi klinis	V0 + negasi dipertahankan + istilah majemuk + ekspansi singkatan
V2	Klinis + ekspansi	V1 + ekspansi sinonim <i>query-side</i>
V3	Char n-gram	indexing karakter n-gram (3-5)
V4	Klinis + ekspansi + PRF	V2 + <i>pseudo-relevance feedback</i>

Arsitektur

`corpus.jsonl` → `ClinicalNormalizer` → `IrIndexer` (index multi-field) → `IrSearcher` (BM25) → endpoint JSON → UI dropdown.

Implementasi Selesai End-to-End

Mesin IR (Java/Lucene 8) + integrasi OpenMRS, teruji berjalan.



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Komponen	Status	Catatan
Mesin IR (Lucene 8 shaded)	Selesai	IrIndexer, IrSearcher, ClinicalNormalizer, BM25 $k_1=1,2$ $b=0,75$.
Index multi-field	Selesai	content_naive/clinical/ngram, 2.316 dokumen.
Modul OpenMRS	Selesai	IrBm25Service + endpoint JSON + UI + <i>sidebar</i> link.
Query set & qrels	Selesai	50 query, qrels TREC semi-otomatis sadar-sinonim.
Evaluasi metrik	Selesai	P@10, Recall@10, nDCG@10 per varian.

Catatan teknis

OpenMRS membundel Lucene lama (tanpa BM25) $\Rightarrow$ Lucene 8.11.2 di-*shade* (relokasi paket) agar tak bentrok. Modul target 1.11.6 teruji berjalan pada OpenMRS 2.8.8.



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Hasil & Evaluasi



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Hasil Evaluasi: Meningkatkan nDCG

50 query, cutoff K = 10.

Normalisasi Klinis



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Varian	P@10	Recall@10	nDCG@10
V0 · Naive BM25	0,5400	0,4312	0,6832
V1 · Normalisasi klinis	0,5420	0,4284	0,6865
V2 · Klinis + ekspansi	0,5120	0,4192	0,6509
V3 · Char n-gram	0,5280	0,4055	0,6446
V4 · Klinis + ekspansi + PRF	0,5020	0,4139	0,6186

Temuan

V1 > V0 pada P@10 & nDCG@10 $\Rightarrow$ normalisasi klinis memberi perbaikan kecil namun konsisten. V2/V4 turun presisi (trade-off ekspansi), dilaporkan apa adanya.

Verifikasi fungsional

Query "CKD" $\rightarrow$ menemukan "Chronic Kidney Disease - Followup"; "hypertension" $\rightarrow$ menemukan dokumen ber-"HTN".

Sumber: eval/results.txt, dijalankan via EvalRunner

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Demo: Mode "Bandingkan Semua"

Kelima varian berdampingan untuk query yang sama ("hypertension").



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OpenMRS

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IR BM25 Search

Query Variant Limit

Side-by-side comparison

Naive BM25

#	Document	Score	Specialties
1	Nephrology Consultation - 3 (mts-01921)	1.7901042	Consult - History and Phys., Nephrology
2	Consult - Hypertension (mts-02122)	1.7901042	Consult - History and Phys., General Medicine
3	Bariatric Consult - Surgical Weight Loss - 3 (mts-02282)	1.7162453	Bariatrics, Consult - History and Phys.

Normalisasi klinis

#	Document	Score	Specialties
1	Consult - Hypertension (mts-02122)	1.7453039	Consult - History and Phys., General Medicine
2	Nephrology Consultation - 3 (mts-01921)	1.7198044	Consult - History and Phys., Nephrology
3	Bariatric Consult - Surgical Weight Loss - 3 (mts-02282)	1.6698341	Bariatrics, Consult - History and Phys.

Klinis + ekspansi

#	Document	Score	Specialties
1	MCA Aneurysm (mts-01873)	4.860256	Consult - History and Phys., Neurology
2	Trouble Breathing - H&P (mts-01994)	4.0895495	Consult - History and Phys., General Medicine
3	Cerebral Angiogram & MRA (mts-01528)	3.8865619	Neurology, Radiology

Char n-gram

#	Document	Score	Specialties
1	Renal Insufficiency - Consult (mts-01915)	27.864851	Consult - History and Phys., General Medicine, Nephrology
2	Atrial Fibrillation - Consult (mts-02285)	27.83517	Cardiovascular / Pulmonary, Consult - History and Phys.
3	Consult - Hypertension (mts-02122)	27.645432	Consult - History and Phys., General Medicine

Klinis + ekspansi + PRF

#	Document	Score	Specialties
1	MCA Aneurysm (mts-01873)	7.562659	Consult - History and Phys., Neurology
2	Cerebral Angiogram & MRA (mts-01528)	5.877725	Neurology, Radiology
3	MRI C-spine (mts-01395)	5.396678	Neurology, Orthopedic, Radiology

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Tangkapan layar modul ir-bm25 pada OpenMRS Legacy UI.

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Demo: Ekspansi Singkatan

Query "CKD" menjembatani dokumen yang memuat "chronic kidney disease".



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IR BM25 Search

Query Variant Limit

Results (5)

#	Document	Score	Specialties	Snippet
1	Chronic Kidney Disease - Followup (mts-01932)	8.086454	Nephrology	REASON FOR VISIT: , Followup on chronic kidney disease.,HISTORY OF PRESENT ILLNESS: , The patient is a 78-year-old gentiema...
2	Acute Kidney Failure (mts-01935)	7.0030684	Nephrology	REASON FOR VISIT: , Acute kidney failure.,HISTORY OF PRESENT ILLNESS: , The patient is a 68-year-old Korean gentiema...
3	Nephrology Office Visit - 2 (mts-01808)	5.379157	Nephrology, Office Notes	...ILLNESS: , This is a 79-year-old white male who presents for a nephrology followup for his chronic kidney disease secondary to nephrosclerosis and nonfunctioning right kidney. His m...
4	Gen Med Consult - 49 (mts-02940)	4.9281483	Consult - History and Phy., General Medicine	HISTORY OF PRESENT ILLNESS: , This is a 70-year-old female with a past medical history of chronic kidney disease, stage 4; history of diabetes mellitus; diabetic nephropathy; perip...
5	Chronic Kidney Disease Followup (mts-01266)	4.724895	Nephrology, SOAP / Chart / Progress Notes	...LNESS: , This is a followup for this 69-year-old African American gentleman with stage IV chronic kidney disease secondary to polycystic kidney disease. His creatinine has ranged b...

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Tangkapan layar hasil pencarian (variant v2).

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Kendala yang Sudah Ditangani



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Lisensi dataset

Ambigu (CCO di Kaggle vs. tanpa lisensi di mtsamples.com).

Selesai: dilaporkan apa adanya pada laporan, tidak mengklaim satu lisensi bersih.

Ground truth evaluasi

medical_specialty multi-label.

Selesai: qrels disusun terpisah (semi-otomatis, sadar-sinonim), arah manual qrels dicatat.

Pilihan tokenizer

Selesai: lexicon klinis kurasi terbuka (negasi, majemuk, singkatan, sinonim), bukan adaptasi UMLS yang berlisensi.

Apa yang dicapai

- ▶ Mesin IR BM25 dengan lapisan normalisasi + ekspansi klinis, terintegrasi ke OpenMRS sebagai modul.
- ▶ Ablation study 5 varian pada korpus MTSamples (2.316 dokumen).
- ▶ Validasi berbasis data: negasi/majemuk/singkatan muncul di mayoritas dokumen.
- ▶ Hasil: V1 (normalisasi klinis) unggul tipis atas baseline naif; trade-off presisi pada varian ekspansi dilaporkan jujur.
- ▶ Purwarupa pencarian cerdas yang siap dipakai dan dapat diperluas.

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Data terverifikasi, gap jelas, mesin berjalan, dan evaluasi lengkap.

Sistem IR BM25 berjalan di OpenMRS: normalisasi klinis memberi perbaikan, varian ekspansi diukur apa adanya, purwarupa siap dipakai.

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